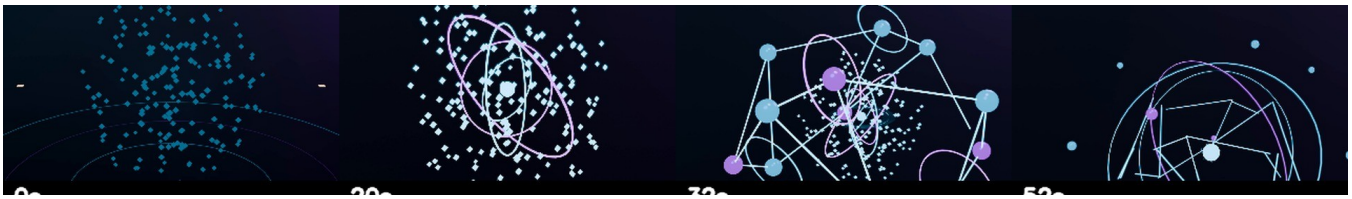
QFG DYNAMICS 1.0.0-alpha

Quick Start & Showcase Guide for Blender 5.2

Unified QFG field, dynamics and structural simulation



Build a field. Make it move. Turn geometry into an interaction graph.
Then close the loop from Atlas to QFG and back again.

READ THIS FIRST

QFG Dynamics is a research workbench. Several sectors are reduced models, proxies, or conditional bridges rather than SI-calibrated first-principles physics. Treat visual beauty, stability, and convergence as useful engineering evidence - not as experimental validation by themselves.

Recommended first run: NumPy CPU | Scalar prototype | 32^3 grid | dt = 0.002

1. What this guide is for

This is not the full reference manual. It is a fast, hands-on tour designed to get a new user from a clean Blender installation to several scenes that show why the plugin is interesting.

- Start with a scene that proves the numerical core works.
- Move into topology, 6-DoF dynamics and thermal/gauge variants.
- Build a BSM-SG / Atlas interaction graph and field bridge.
- Finish with bounded feedback, constraints and a reproducible blind-relaxation workflow.
- Bake results to the Blender timeline so they can be inspected, presented and rendered.

THE 10-MINUTE FAST LANE

If you only want to see the plugin come alive: install it, open QFG 5.2, choose Starting Scenes -> Two-node collision, initialize the engine, run several steps, update the visualization, then try Native Hero Showcase in Preview quality.

Where things live

Location	Purpose
Scene-level tools	3D Viewport -> Sidebar (N) -> QFG 5.2
Object-level tools	Properties -> Object -> QFG Actor 5.2
Core controls	QFG Dynamics 1.0.0 Alpha + Numerical Core
Timeline tools	Visualization and Timeline
Dynamics debug	6-DoF Debug Overlay
Atlas pipeline	Atlas Direct / BSM-SG Geometry
Export	Diagnostics and Export + Export Atlas Report

2. Install and verify

1. Use Blender 5.2 LTS. Do not unpack the extension ZIP.
2. Open Edit -> Preferences -> Extensions.
3. Use the menu in the upper-right corner and choose Install from Disk.
4. Select qfg_dynamics_blender_5_2_v1.0.0-alpha.zip and enable QFG Lab 5.2 / QFG Dynamics 1.0.0 Alpha.
5. Return to the 3D Viewport, press N, and open the QFG 5.2 tab.
6. For the first run, choose NumPy CPU. PyTorch/CUDA is optional and should come later.

GOOD BASELINE

Backend = NumPy CPU; Field = Scalar prototype; Grid = 32^3; Domain length = 16; dt = 0.002; Steps per action = 10. This is the safest first configuration for checking installation and UI registration.

SCENE 1 | 3-5 min

Two-node collision - the first five minutes

Prove that the core solver, scene registration and visualization pipeline are alive.

Why this scene matters

This preset is the quickest sanity check. It exercises QFG field sources, numerical stepping, scene registration and the visualization pipeline without asking you to understand the entire engine first.

1. Open QFG 5.2 -> Numerical Core. Set Backend = NumPy CPU.
2. Set Field = Scalar prototype, Grid = 32^3 Standard, Domain length = 16 and dt = 0.002.
3. Open Starting Scenes and choose Two-node collision.
4. Build the preset, then click Initialize QFG Engine.
5. Click Run QFG Steps several times. Each click advances Steps per action numerical steps.
6. Open Visualization and Timeline and click Update Visualization.
7. Inspect QFG_Field_Points, the central density slice, and Diagnostics and Export.

WHAT TO LOOK FOR

A visible field structure, changing density, finite diagnostics and a scene that updates without exceptions. If the visualization is empty, lower Density threshold and raise Maximum points.

TRY THIS NEXT

Repeat with dt/2, then with 48^3 or 64^3. A result worth interpreting should not change dramatically when the discretization is modestly refined.

SCENE 2 | 5-8 min

Hopf ring and vortex - make topology visible

Switch from a simple scalar scene to reduced spinor/topology structures that immediately look different.

A. Hopf-ring prototype

1. In Starting Scenes choose Hopf-ring prototype.
2. Set the field mode to the reduced two-component Spinor prototype if the preset does not do so automatically.
3. Initialize the QFG Engine and run several actions.
4. Open Field and Topology. Enable the topology / Skyrme sector only if it is not already enabled by the preset.
5. Run Compute Topology and note Hopf proxy, Skyrme proxy and Orientation variance.
6. Update Visualization and orbit the camera around the ring-like structure.

WHY IT HOOKS PEOPLE

The scene changes the mental model from “a cloud of values” to a textured, oriented field. The topology metrics are reduced numerical proxies, but the 3D structure is a compelling way to explore orientation and winding.

B. Vortex prototype

1. Reset or build a clean Vortex prototype from Starting Scenes.
2. Initialize the engine and run steps.
3. Update the visualization and inspect the phase-winding / spinor texture.
4. Compare the visual structure and topology diagnostics with the Hopf-ring run.

DO NOT OVER-CLAIM

Hopf proxy, Skyrme proxy and texture metrics are comparative diagnostics. They are not automatically proven topological invariants without convergence and boundary analysis.

SCENE 3 | 8-12 min

6-DoF field-coupled bodies - turn QFG into motion

Create Blender objects that translate, rotate, collide and couple to the field.

Build two dynamic bodies

1. Create two simple meshes in Blender - UV spheres or boxes are ideal for a first test.
2. Select the first object. Open Properties -> Object -> QFG Actor 5.2 and enable QFG actor.
3. Set Actor type = QFG dynamic body.
4. Give it a modest linear velocity, Body mass, collision radius or half-extents, and non-zero field coupling.
5. Repeat for the second object with an opposing velocity.
6. Open Dynamics Runtime. Enable body collisions, set Dynamics substeps to a modest value, and enable Deterministic replay with a fixed World seed.
7. Initialize the engine and run steps. Watch the bodies move, collide and respond to the field.
8. Open the 6-DoF Debug Overlay and enable the HUD, force vectors, torque vectors and contacts.

WHAT MAKES THIS DIFFERENT

Blender is not just drawing a result here. The scene becomes an interactive front-end to a reduced rigid-body runtime: translation, rotation, contacts, field coupling, thermal coupling, forces and torques can be inspected together.

Object setting	What it controls
Linear / angular velocity	Initial motion
Body mass	Reduced mass parameter
Collision shape	Sphere or oriented box
Field / thermal / torque coupling	Per-body coupling to engine sectors
Drag	Numerical damping for motion
Restitution / friction	Contact response
Kinematic	Externally controlled body rather than freely integrated body

AAA DEMO TIP

After the motion looks good, bake it to the timeline and render it with a slow camera move, glow and motion blur. The engine data stays the driver; the presentation layer can be as cinematic as you like.

SCENE 4 | 6-10 min

Thermal collision and gauge + gravity - layer the physics proxies

Use ready-made scenes to see how additional sectors alter the same visual language.

Thermal collision

1. Build Starting Scenes -> Thermal collision.
2. Open Thermal and HRM. Note Heat conduction, Collision heating, Critical temperature and Temperature cap.
3. Initialize, run steps and update the visualization.
4. Compare Max T and the dynamics before and after the collision.

THE HOOK

The collision is no longer only mechanical. The scene carries a temperature field and a collision-heating proxy, so one event becomes visible in several linked diagnostic channels.

Gauge + gravity

1. Build Starting Scenes -> Gauge + gravity.
2. Open Gauge, Gravity and Noise. Inspect the reduced U(1) and weak-field gravity controls.
3. Initialize, run steps and compare Gauge E, Gravity E and the field visualization.
4. Add a small noise amplitude and fixed seed to test whether the qualitative result survives a controlled perturbation.

SCIENTIFIC STATUS

The gauge sector is a reduced A/E-field model and the gravity sector is a Poisson-like weak-field proxy. They are research tools, not a full Maxwell or Einstein solver.

SCENE 5 | 10-15 min

Atlas Direct - turn geometry into a graph

Tag source geometry, extract structural quantities, build J_ij, orbital planes and an integrated envelope.

B1 - tag and extract

1. Create or import a small set of objects representing the structural sites you want to study.
2. For each relevant object, enable QFG Actor 5.2 and choose Atlas nucleon when appropriate.
3. In Atlas Direct / BSM-SG Geometry choose the nuclear center rule and review the source geometry.
4. Tag objects as CORE, VALENCE, BOUND or EQUATORIAL as required by your model.
5. Use explicit SOURCE provenance for reviewed source geometry. Use DERIVED only for deterministic quantities inferred from SOURCE inputs.
6. Run Extract Atlas Geometry. Inspect pair angles, mean angle, spread, min/max and the C3v h/r proxy when relevant.

B2 - build the interaction picture

1. Choose an interaction topology. Source star is the minimal topology for the current Group-15 workflow; Valence pairs and All tagged are useful diagnostics.
2. Build the J_ij interaction graph.
3. Start with Topology only if you want a pure connectivity diagnostic.
4. Only then experiment with orientation-overlap or softened 1/r^3 proxy kernels.
5. Generate orbital planes using explicit local-axis rules. Avoid measuring scientific angles from the camera projection.
6. Build the integrated Bohr envelope for a geometric overview of the structure.

WHY THIS IS A GREAT SHOWCASE

In one viewport you can show source geometry, semantic roles, an interaction graph, orientation planes and an envelope. The plugin turns a Blender scene into a structured computational object rather than a decorative model.

PROVENANCE LEGEND

SOURCE = reviewed input; DERIVED = deterministic result from source inputs; CONDITIONAL = model bridge or hypothesis; CALIBRATED = uses a reference to fix a parameter; EXPERIMENTAL = external measured/reference value; REJECTED = audit/history only.

SCENE 6 | 10-15 min

Close the loop - Atlas -> QFG -> Atlas

Rasterize structural sites into a QFG field, then let the sampled field feed back into geometry.

B3 - Atlas to QFG field bridge

1. Enable Atlas field bridge.
2. For a first run choose Field source scope = SOURCE only.
3. Choose a role scope appropriate to the scene - for example Valence + equatorial.
4. Use the Gaussian kernel first. It is the recommended finite numerical source.
5. Keep Vector/spin scale at 0 until you actually need an orientation bridge.
6. Apply Atlas Field and inspect source count, V min/max/RMS and |B| max/RMS.
7. Update the QFG visualization and confirm that the field follows the tagged structural sites.

TRY THIS

Move one SOURCE object slightly, re-rasterize, and watch the field change. This is the moment where Blender geometry becomes an input to the numerical engine.

B4 - QFG back to Atlas

1. Enable B4 bidirectional feedback.
2. Capture Anchors before relaxation so the reviewed geometry can be restored.
3. Choose a Torque target: Density gradient, Spin texture or Hybrid.
4. Use small force/torque scales and retain the safety clamps for maximum translation and rotation per cycle.
5. Run Apply Feedback Once and inspect the direction and magnitude before committing to a longer relaxation.
6. Rebuild B3 after feedback if you want the field to track the changing geometry.

IMPORTANT

B4 is a conditional feedback law. Its force/work quantities are audit proxies, not SI-calibrated newtons or joules.

SCENE 7 | 10-20 min

B5 + B6 - bounded relaxation and a reproducible blind run

Keep the geometry admissible, converge without reading the holdout, freeze the prediction and only then compare.

B5 - preserve admissible structure

1. Enable Geometry constraints.
2. Capture connectivity from the source structure - for example Manual neighbours, Parent-child, or B2 Source star.
3. Use pair-distance, shell, angular-cone and symmetry stiffness only when they are justified by the model definition.
4. For a C3 valence set, give the three movable arms the same symmetry group. The collective polar tilt should remain free.
5. Apply constraints and watch the correction size. Large corrections on every iteration are a warning that the field feedback and admissible geometry disagree.

B6 - blind relaxation

1. Keep the experimental holdout hidden.
2. Use Auto-initialize QFG, Reset to B4 anchors and Recapture B5 constraints for a clean protocol.
3. Run Blind Relaxation.
4. Watch convergence window, angle range, angle drift, C3 spread and any optional B5 correction gate.
5. If converged = True, Freeze Prediction.
6. Record the SHA-256 protocol fingerprint.
7. Only now use Reveal Experimental Holdout and generate the residual report.
8. Export Atlas Report and preserve the .blend file plus the numerical session export.

THE POINT OF B6

B6 does not invent a new force law. It orchestrates a blind, auditable run so the stopping rule cannot read the experimental target. The protocol hash becomes a compact identity for the numerical setup.

DO NOT FREEZE A BAD RUN

If the angle still oscillates, safety clipping is constant, B5 corrections remain large, or the result changes dramatically with dt/grid, treat the run as unresolved rather than forcing a conclusion.

3. Bonus: build something worth showing

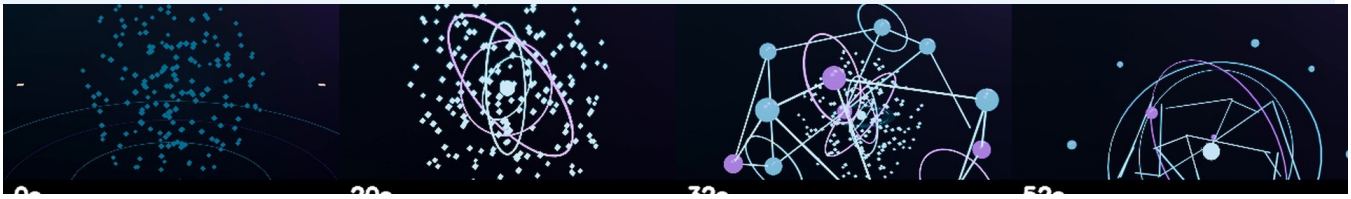
Once a numerical scene is behaving well, the same data can become a presentation scene. The plugin already includes Starting Scenes and a Native Hero Showcase workflow, and baked engine results can be rendered with ordinary Blender tools.

Native Hero Showcase

1. Open Starting Scenes -> Native Hero Showcase.
2. Use Quality = Preview while experimenting.
3. Choose the showcase mode you want to highlight - Collision, Hopf, Vortex, HRM cavity, Thermal, Gauge + gravity or Plasma boundary.
4. Build the scene. For a timeline sequence, use Build + Bake so the heavy numerical work is computed before playback.
5. Inspect the hero frame, then use Render Still or Blender animation rendering for presentation output.

PRESENTATION RULE

Separate computed content from decoration. Density, field evolution, thermal values, topology proxies and dynamic trajectories come from the engine. Lighting, rings, atmosphere, lens effects and camera moves are presentation layers.



Example cinematic progression: field awakening -> coherent structure -> dynamics -> Atlas/J_ij -> integrated hero shot.

4. Bake, inspect and render

1. Set Bake start / end in Visualization and Timeline.
2. Use Bake QFG Timeline. This is preferred to running a heavy 3D FFT solve inside a frame-change playback handler.
3. Scrub the timeline and confirm that the cached result is stable and visually correct.
4. Use the 6-DoF Debug Overlay while diagnosing, but disable it for a clean final render unless the HUD is part of the presentation.
5. Render a still or animation with Blender Eevee/Cycles. Use motion blur and glow only after the underlying cached states are correct.

SAVE THE AUDIT TRAIL

For every important run keep: Blender file, session export, Atlas report, plugin version, backend/grid/dt, deterministic seed where relevant, and the B6 protocol hash for blind-relaxation runs.

5. Quick reference - what to try next

Goal	Fast path
Check that the plugin works	Two-node collision -> Initialize -> Run Steps -> Update Visualization
Create a field source	Object -> QFG Actor 5.2 -> Gaussian / Hopf / Vortex / Plane wave
Run rigid-body dynamics	QFG dynamic body + Dynamics Runtime + 6-DoF Debug Overlay
Explore topology	Spinor mode -> Topology enabled -> Compute Topology
Add thermal behavior	Thermal collision + Thermal and HRM
Try reduced gauge/gravity	Starting Scenes -> Gauge + gravity
Analyze Atlas geometry	Atlas Direct -> Tag -> Extract Atlas Geometry
Build an interaction graph	B2 -> Source star -> Build Interaction Graph
Feed Atlas into QFG	B3 -> Enable -> Apply Atlas Field
Feed QFG back into geometry	B4 -> Capture Anchors -> Apply Feedback
Keep geometry admissible	B5 -> Capture Constraints -> Apply Constraints
Run a blind prediction	B6 -> Run -> Converge -> Freeze -> Reveal
Preserve results	Export Atlas Report + Export Session + .blend file

6. Troubleshooting

Symptom	First check
Plugin does not appear	Check Blender 5.2, Extensions enabled, and the correct ZIP package.
Torch/CUDA fails	Use NumPy CPU for the control run; PyTorch is optional and not installed automatically.
Run becomes unstable	Reduce dt; reduce coupling/feedback; check safety clips; use a smaller grid for diagnosis.
Visualization is empty	Lower Density threshold; increase Maximum points; click Update Visualization.
Atlas angles look wrong	Check the nuclear center, VALENCE tags and direction

	rules; never measure scientific angles from the camera view.
B3 field is zero	Check bridge enable, source/role scope, per-object B3 field source and scalar/vector scales.
B4 does not move objects	Check feedback enable, anchors, per-object mobility/coupling and position/rotation locks.
B6 will not converge	Do not force Freeze. Check B4/B5 conflict, tolerances, dt/grid and oscillating trajectories.

7. Where to go after this guide

The plugin is deliberately broader than a single demo. Once these scenes work, the next step is to use the full documentation for the specialized modules: RF/HRM/plasma ROM, cavity/MEMS, Yb quantum-control tools, Molecular Creator, advanced provenance, convergence testing and the full Atlas Direct B1-B6 workflow.

A GOOD HABIT

For any result you care about: run a Preview configuration first, export it, repeat with dt/2, repeat at a higher grid, and only then interpret the pattern. Reproducibility is part of the feature set, not paperwork after the fact.

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